



Demystifying AI Security

Applying SAIF (Google's
Secure AI Framework) to
Eliminate Hidden Risk

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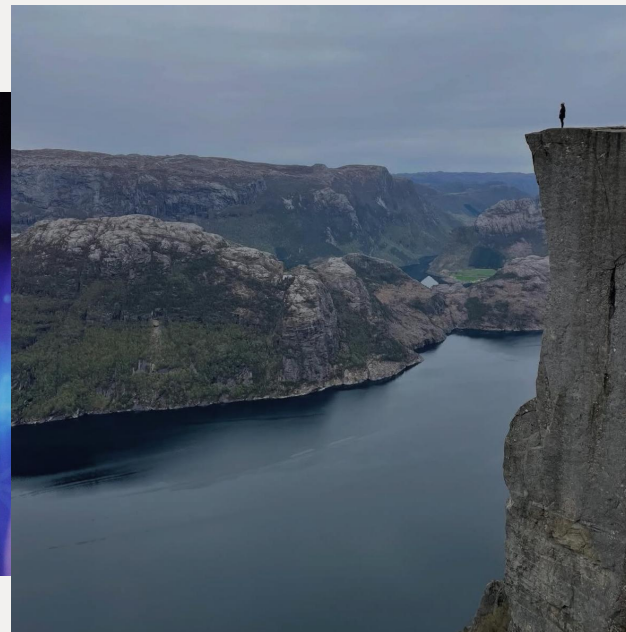
Innovator : SIH Winner, UIA

Public speaker

Athlete at heart : 🏃🏆🏀🏈🏸



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AGENDA

- 1 Why AI security is critical in today's data-driven world
- 2 A quick intro to Google's Secure AI Framework (SAIF)
- 3 Applying SAIF Across Dimensions
- 4 Eliminating Unknown Risks
- 5 Operationalizing AI Security
- 6 Takeaways & Call to Action

Why AI security is critical in today's data-driven world ?

- AI adoption is exploding
- Cloud risks fly under the radar
- Organizations don't know they have sensitive data lurking in their environment
- This creates a treasure trove for attackers.
- PII, Sensitive info creating high-value attack targets



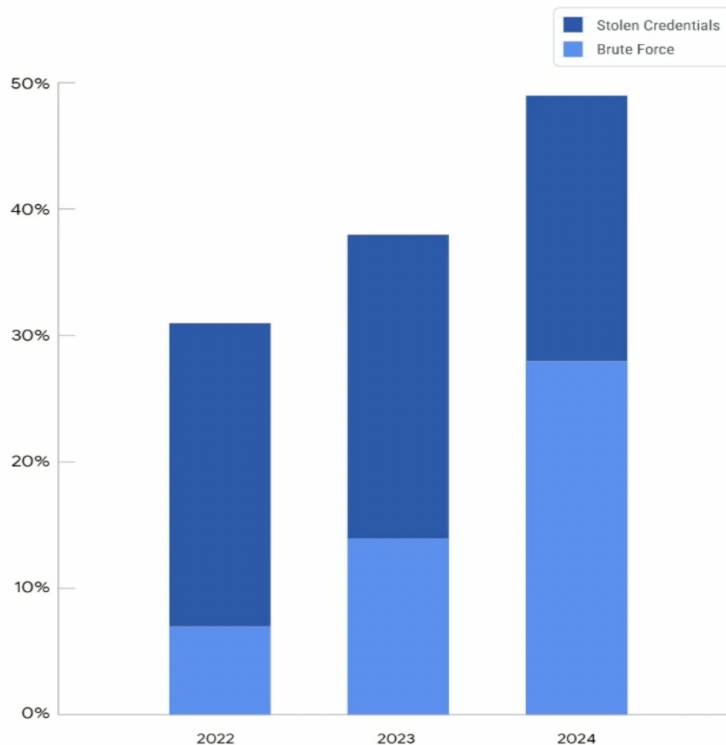
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**Compromised Identity is
the root cause**

50%

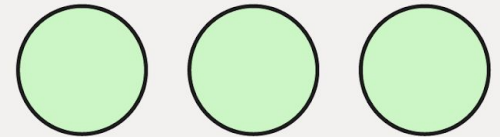
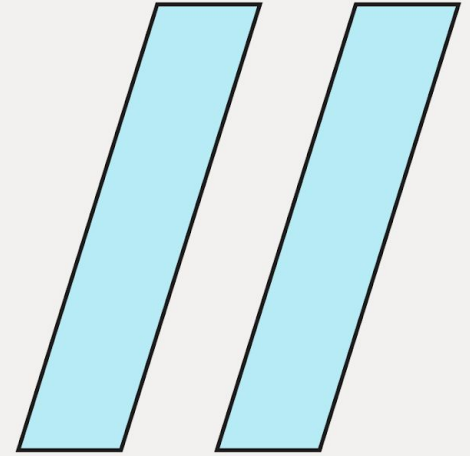
of the ransomware intrusions
responded to, were a result of
compromised identity

Compromised Credential Usage for
Initial Access in Ransomware Incidents





There is no security
without secure
identity.



Google: a Leader in securing enterprise data



Google differentiates with data threat and risk visibility, access controls, masking, encryption, and addressing supplier risk. It is superior for privacy (including confidential computing), information governance, and AI security and governance use cases.

The Forrester Wave™: Data Security Platforms, Q1 2025

FORRESTER

POLLER 1

Forrester Wave™: Data Security Platforms, Q1 2025

The Forrester Wave™: Data Security Platforms, Q1 2025
The 10 Providers That Matter Most And How They Stack Up

THE FORRESTER WAVE™

Data Security Platforms

Q1 2025

Customer feedback*



*A halo indicates above average customer feedback. A double halo indicates that the vendor is a Customer Favorite.

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AI and Security

AI and security can largely be organized into three buckets:

01

Securing the
use of AI

02

Using AI to secure
other systems

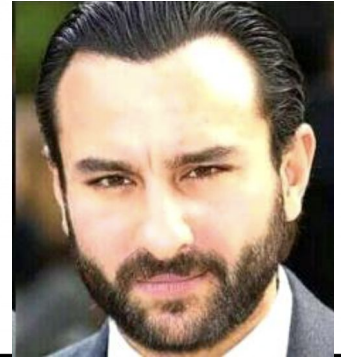
03

Countering the
attacker armed with AI



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What is SAIF?



Secure AI Framework Approach



What's SAIF?

- SAIF (Secure AI Framework) is a security **framework** published by Google that provides a structured approach to **building, deploying, and maintaining safe, secure, and trustworthy AI systems.**



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Why SAIF matters?

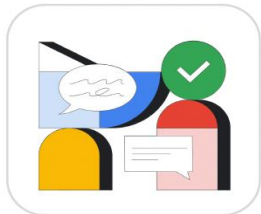
- As AI adoption rapidly accelerates -> poor security practices scale
- SAIF reduces this risk by enforcing security-by-design.
- SAIF prevents threats like data leakage, poisoning, jailbreaks.



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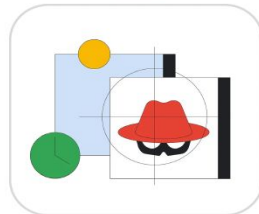
SAIF 6 pillars

01



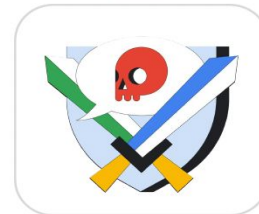
Expand strong security foundations to the AI ecosystem

02



Extend detection and response to bring AI into an organization's threat universe

03



Automate defenses to keep pace with existing and new threats

04



Harmonize platform level controls to ensure consistent security across the organization

05



Adapt controls to adjust mitigations and create faster feedback loops for AI deployment

06



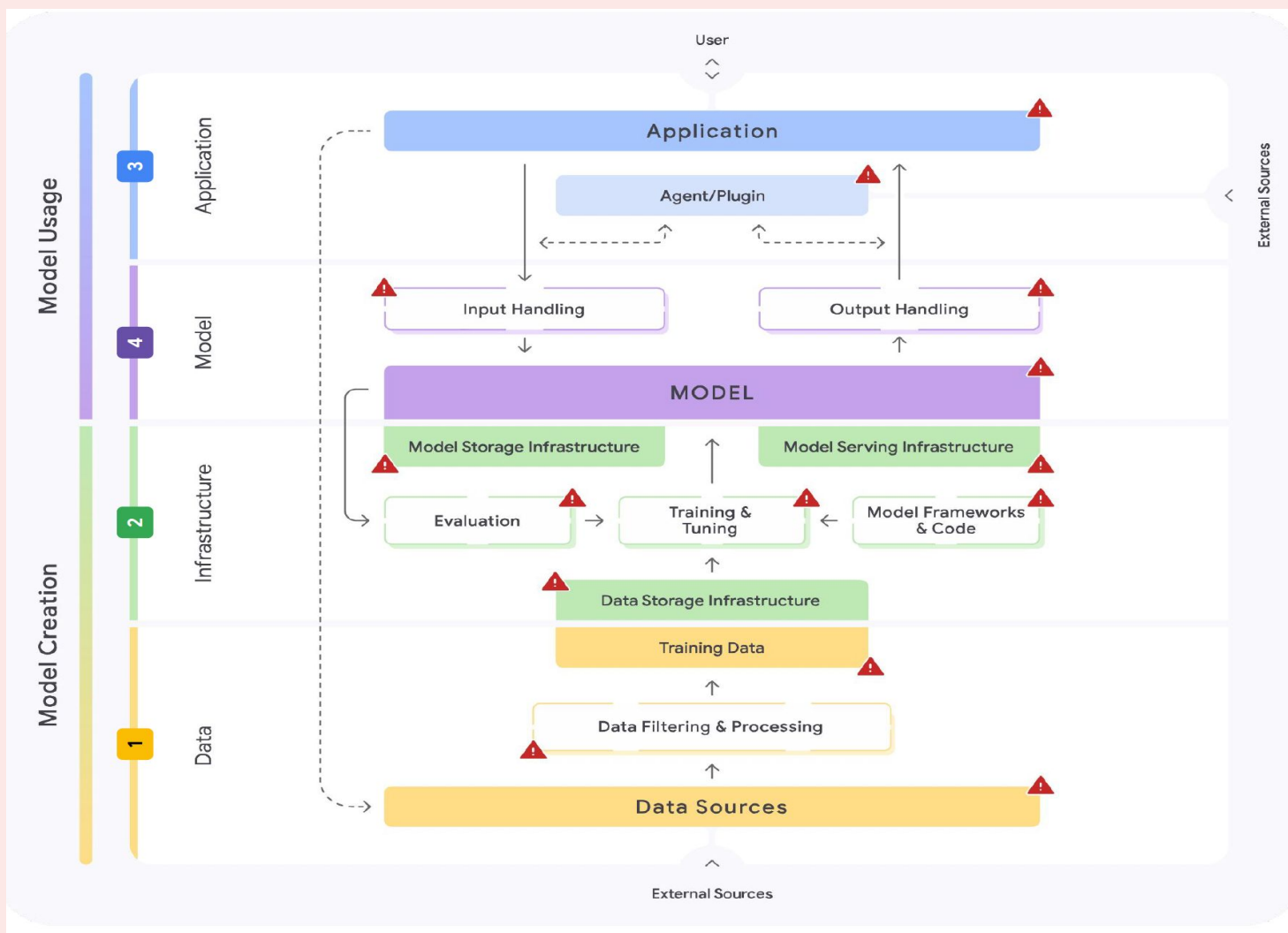
Contextualize AI system risks in surrounding business processes

Overlay SAIF principles on Model Lifecycle

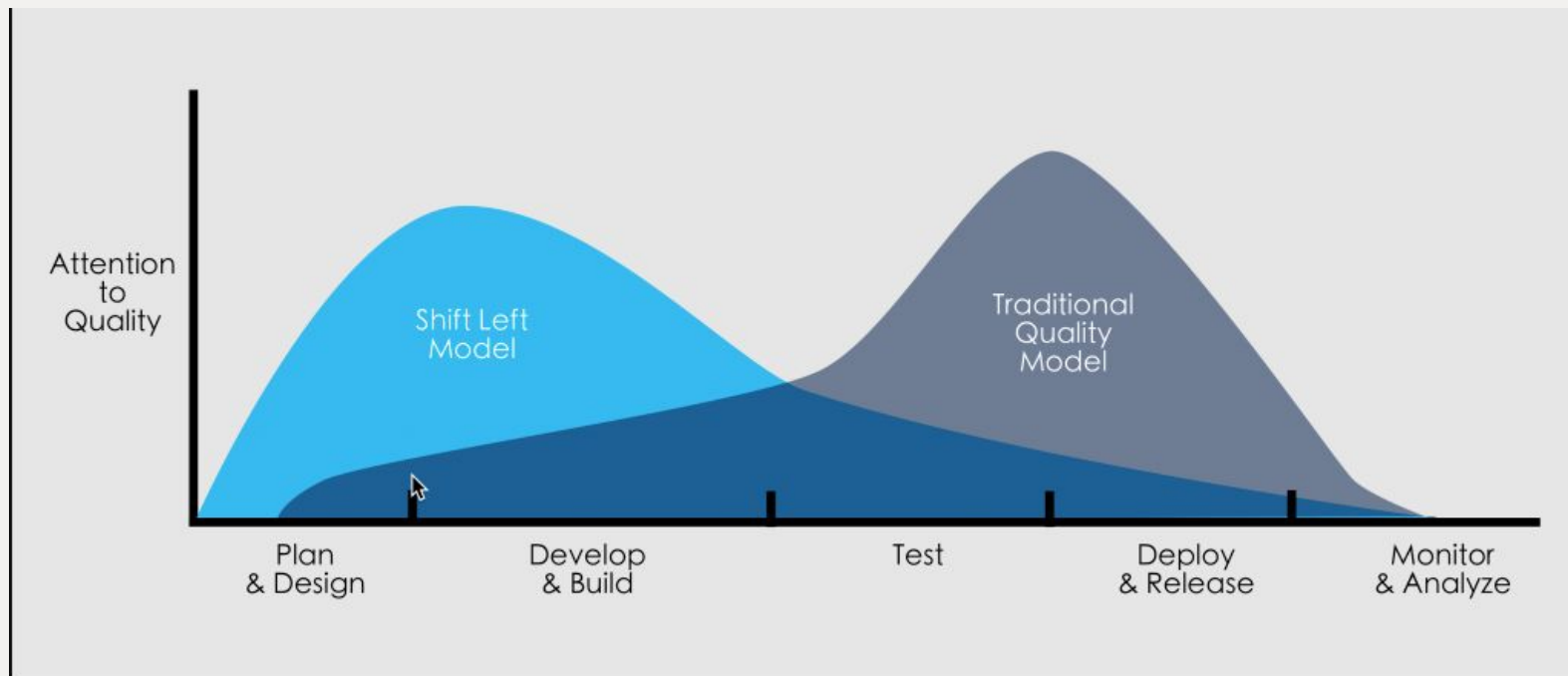
Discovery and problem definition	● Identify security and compliance requirements early. ● Evaluate data sensitivity and ethical risks before choosing an AI approach.
Data Collection and Preparation	● Collect the data that you will need to train your AI model. ● Assess datasets for vulnerabilities (bias, poisoning risk) ● Analyze & label the data (masking)
Model design and development	● Design and develop the AI model that will address the business opportunity that you have defined. ● Design mechanisms to prevent prompt injection, data leakage. ● Define authenticated access and authorization pathways. ● Audit model performance. ● Red-team models during development to find vulnerabilities.

Overlay SAIF principles on Model Lifecycle

Model training, fine-tuning, and testing	• Secure training infrastructure (identity, environment isolation). • Data governance to make sure training data is not tampered. • Monitor training workflows for anomalies. • Ensure testing includes misuse and edge-case scenarios.
Model deployment and integration with end-product	• Apply runtime defenses (rate limiting, monitoring, filtering). • Ensure minimal access to sensitive data. • Establish policies for version control and rollback.
Model behavior monitoring and adjustment	• Continuously log usage and flag abnormal behavior. • Monitor for drift, bias, jailbreaks, or new attack vectors. • Reassess risks regularly and re-train(feedback loops). • Validate performance + safety metrics over time.



Shift Left



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Shift Left?

- In the context of AI development, shifting left ensures:
 - Data used for training is verified and safe
 - Models are designed to resist attacks (e.g., prompt injection)
 - Access and deployment controls are planned early



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Relevant Questions in the Model Lifecycle

Discovery and problem definition	• What type of data will the model need? • Do you have permission to use the data for this purpose? • What will the model do? That is, what is the use case? • Will it be internal or external facing? • Where will the model be hosted? On prem, CSP, or SAAS offering?
Data Collection and Preparation	• How will the organization ensure the quality and integrity of the data collected? • How will the data be anonymized or de-identified to protect privacy? • What measures will be implemented to protect the data from tampering?
Model design and development	• What kind of access control is used to restrict access to training environment? • How will the organization ensure that the model is used in a safe and responsible manner? • How will you deal with vulnerabilities in your chosen ML framework?

Relevant Questions in the Model Lifecycle

Model training, fine-tuning, and testing	• What stress tests will be used to evaluate the model's robustness? • How will the model's operation be monitored and evaluated? • What processes will be in place to track and document changes made to the model during fine-tuning?
Model deployment and integration with end-product	• How will the organization ensure the AI model is deployed securely into the production environment? • How will the integration of the AI model with the end product be secured, especially regarding data flow and API security?
Model behavior/outcome monitoring and adjustment	• What monitoring and logging mechanisms will be implemented to track the model's performance? • What processes will be in place to manage and respond to security incidents discovered after deployment?

SAIF **down**,
lifecycle **right**

Opportunity
discovery and
problem
definition

Data collection
and preparation

Model design
and development

Model training,
fine-tuning, and
testing

Model
deployment
and integration
with end-product

Model
behavior/outcome
monitoring and
adjustment

1

Expand strong
security foundations
to the AI ecosystem
(SAIF1)



2

Extend detection and
response to bring AI
into an organization's
threat universe (SAIF2)



3

Automate defenses to
keep pace with
existing and new
threats (SAIF3)



4

Harmonize platform
level controls to
ensure consistent
security across the
organization (SAIF4)



5

Adapt controls to
adjust mitigations and
create faster
feedback loops for AI
deployment (SAIF5)

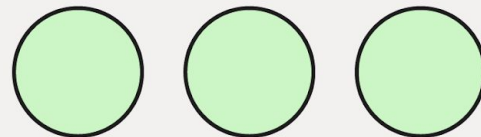
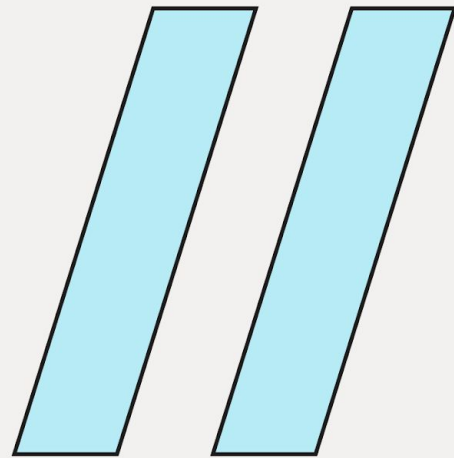


6

Contextualize AI
system risks in
surrounding business
processes (SAIF6)



AI/ML and gen-AI workloads are fueled by data and data context. Understanding and protecting this sensitive, enterprise-specific data is critical to ensuring successful deployments and proper use.



Sensitive data can find its way into model training or tuning data

Google Cloud | dlpsbqrun1-admin2s | vertex

Vertex AI

Tools

- Dashboard
- Model Garden
- Pipelines

Notebooks

- Colab Enterprise
- Workbench

Vertex AI Studio

- Overview
- Freeform

sec_test3 | sec_test3_tcn

Import | **Browse** | Analyze | Lineage

Labels

All		5
Labeled		0
Unlabeled		5
Filter		+

Add new label

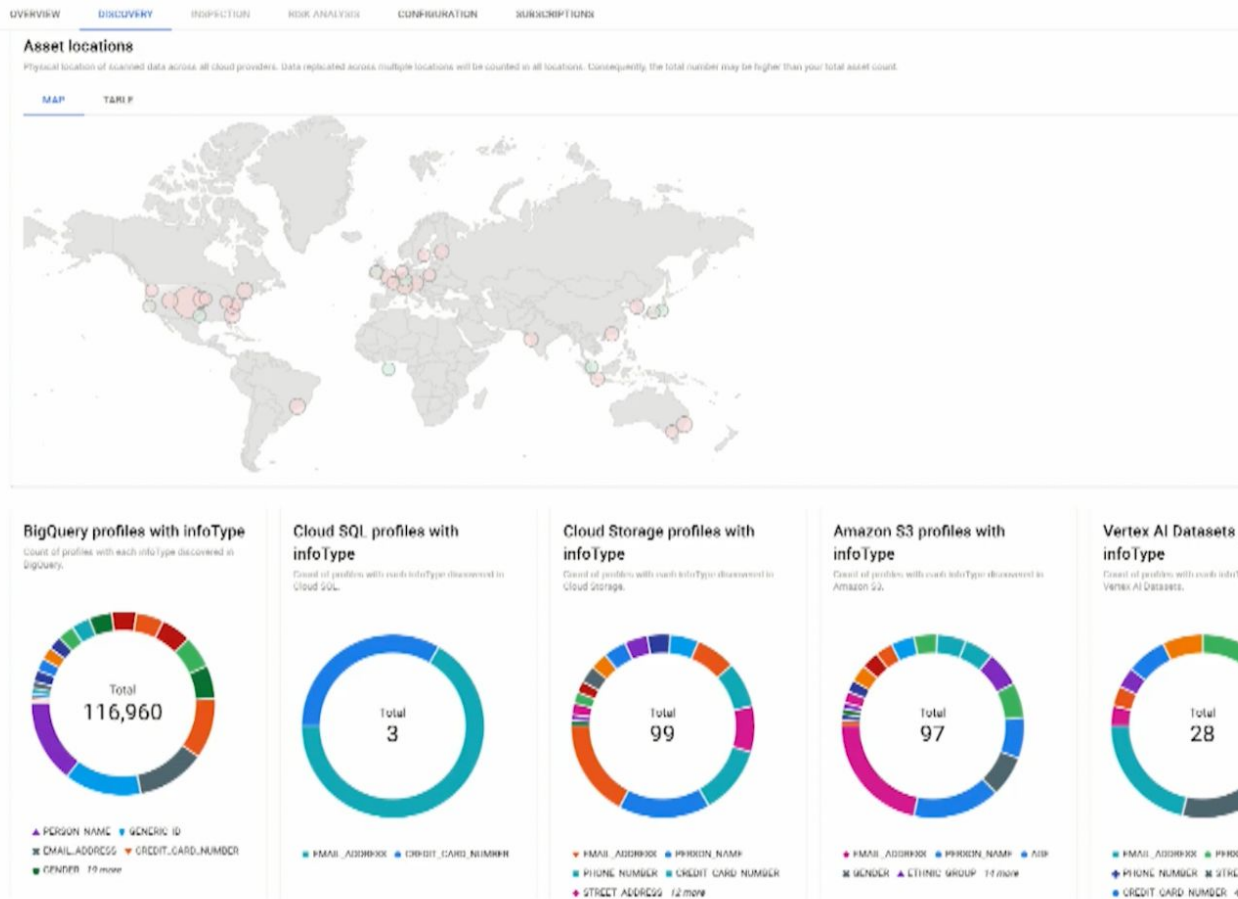
Images

Filter Enter label or property name

	Text	Labels
☐	{ "type": "service_account", "project_id": "fake-abcdef", "private_key_id": "95a15e7...	
☐	"FIREBASE_API_KEY": "AlzaSybMawzEX4-dgZyN72xvDVF_Ya270ka2bcW"	
☐	script async="" src="https://maps.googleapis.com/maps/api/js?key=AlzaSybM...	
☐	GCP_API_KEY = 'AlzaSybMawzEX4-dgZyN72xvDVF_Ya270ka2bcW'	
☐	my password is D3HJKH2sadf832934hj2	

First, remove the unknowns with continuous data discovery.

Go enable Sensitive Data Protection across your org



Second, now that you have insights, make sure data is in the right place.

Option 1: Mask out data that you don't need, but keep the context

Input

Hey Bill, I have a question about my Bill from January

Output

Hey [PERSON_NAME], I have a question about my Bill from January

Input

[Customer] Oh my card, it is 4111-1111-1111-1111 is that what you need?

Output

[Customer] Oh my card, it is [CREDIT_CARD_NUMBER] is that what you need?



Option 2

Completely remove or block unwanted data types

Secrets like

- AUTH_TOKEN
- AWS_CREDS
- GCP_CREDS
- ENCRYPTION_KEY

Sensitive Documents like

- Source Code
- Resume
- Financial Context
- Medical Context
- Political / Religious,etc



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Automate Filtering with De-Identification

The diagram illustrates the process of automating filtering with de-identification. It begins with a file explorer interface showing a directory structure with files like 'file_01.csv', 'file_02.csv', etc. A large black arrow points from this interface to a stack of documents representing redacted data. The documents show 'Redacted Image' and 'Contact Details on File' with redacted fields. Finally, a table titled 'Tokenized and Redacted Data' shows the resulting data with columns for ID, Job Title, Phone, and Comments, where sensitive information has been replaced with tokens.

ID	Job Title	Phone	Comments
359740	Senior Engineer	807-968-0678	Please email them at Jane@imadethisup.com
981587	VP Engineer	713-910-6707	none
394091	Lowyer	692-390-4146	Updated phone to: 692-398-4146
980941	Senior Ops Manager	294-467-5508	none
490456	Junior Ops Manager	791-954-3281	Tried to verify account with their SSN 222-84-5565

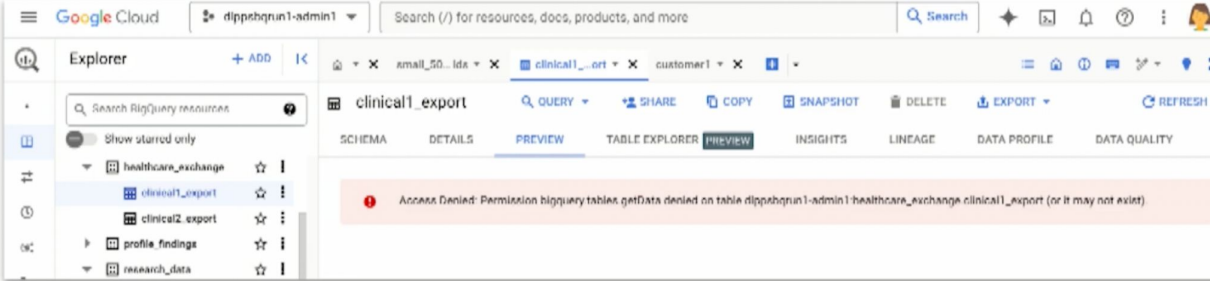
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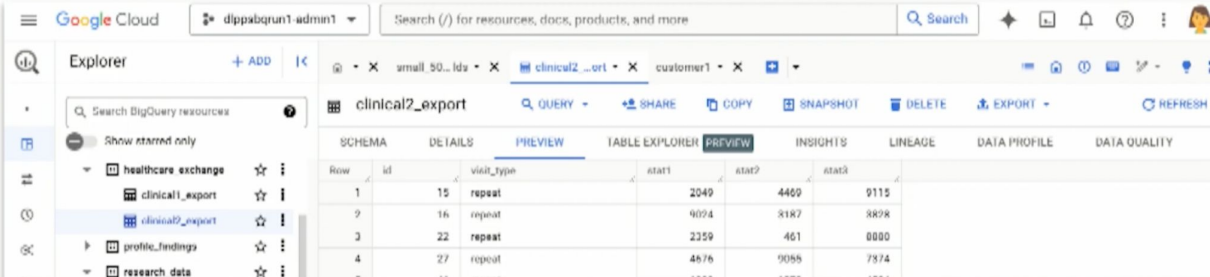
3. Make sure data has right Access Control : make use of tag based conditional access

Highly Sensitive



The screenshot shows the Google Cloud BigQuery Explorer interface. The left sidebar lists the 'healthcare_exchange' dataset with tables 'clinical1_export', 'clinical2_export', 'profile_findings', and 'research_data'. The main panel shows the 'clinical1_export' table with tabs for SCHEMA, DETAILS, PREVIEW, TABLE EXPLORER, INSIGHTS, LINEAGE, DATA PROFILE, and DATA QUALITY. The PREVIEW tab is active, displaying an 'Access Denied' error message: 'Access Denied: Permission bigquery.tables.getData denied on table dippsbqrn1-admin1/healthcare_exchange/clinical1_export (or it may not exist)'.

Not Sensitive

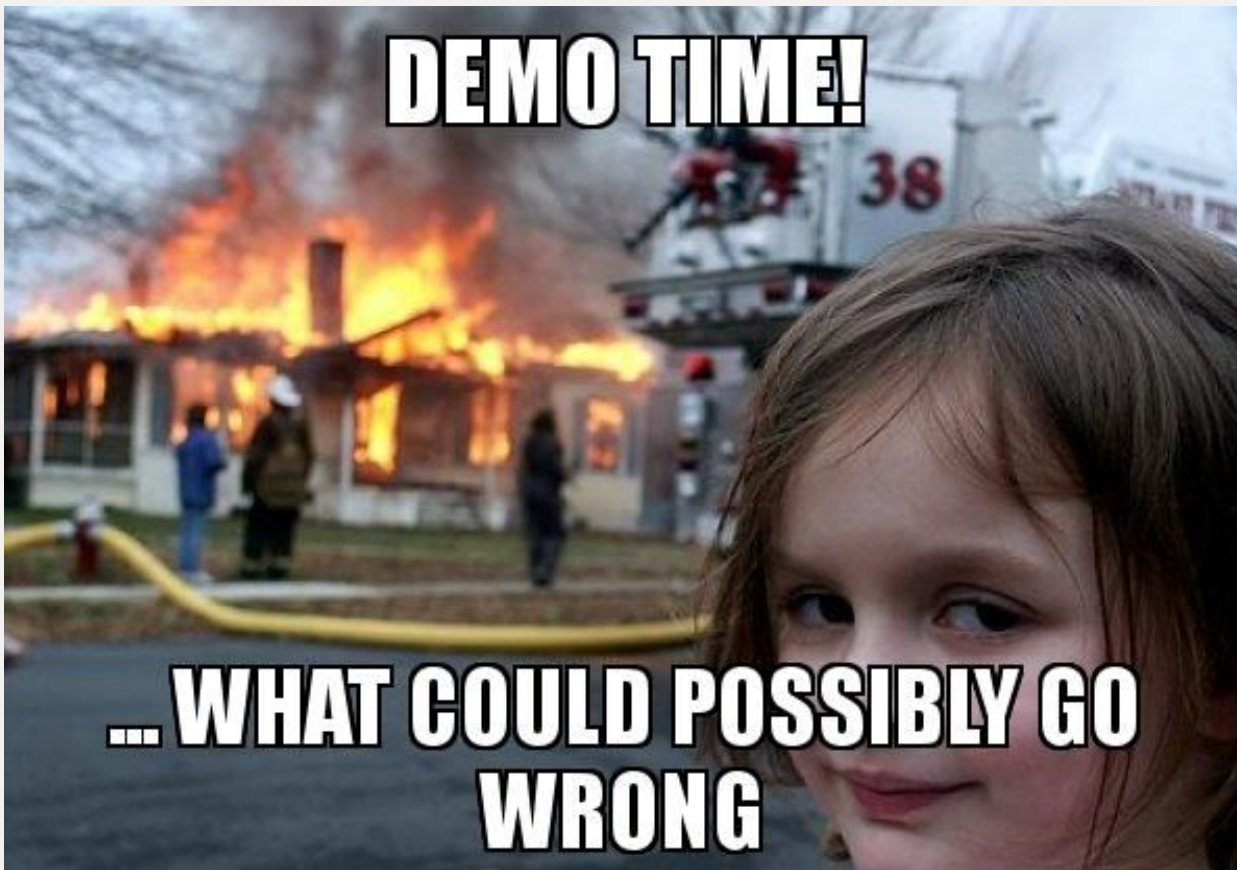


The screenshot shows the Google Cloud BigQuery Explorer interface. The left sidebar lists the 'healthcare_exchange' dataset with tables 'clinical1_export', 'clinical2_export', 'profile_findings', and 'research_data'. The main panel shows the 'clinical2_export' table with tabs for SCHEMA, DETAILS, PREVIEW, TABLE EXPLORER, INSIGHTS, LINEAGE, DATA PROFILE, and DATA QUALITY. The PREVIEW tab is active, displaying a table of data.

Row	id	visit_type	stat1	stat2	stat3
1	15	repeat	2049	4460	9115
2	16	repeat	9094	8187	8898
3	22	repeat	2359	461	8880
4	27	repeat	4876	9088	7874
5	46	repeat	6885	1570	8884

DEMO TIME!

**...WHAT COULD POSSIBLY GO
WRONG**



Automated Masking and Filtering

Tag: Highly Sensitive Data



DENY

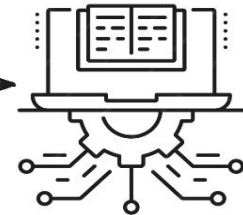


Vertex AI Service Account
& AI Data Engineer

Tag: Low Sensitive Data



ALLOW



TRAINING DATA

Model Tuning and
Training RAG

Demo Overview

We will use:

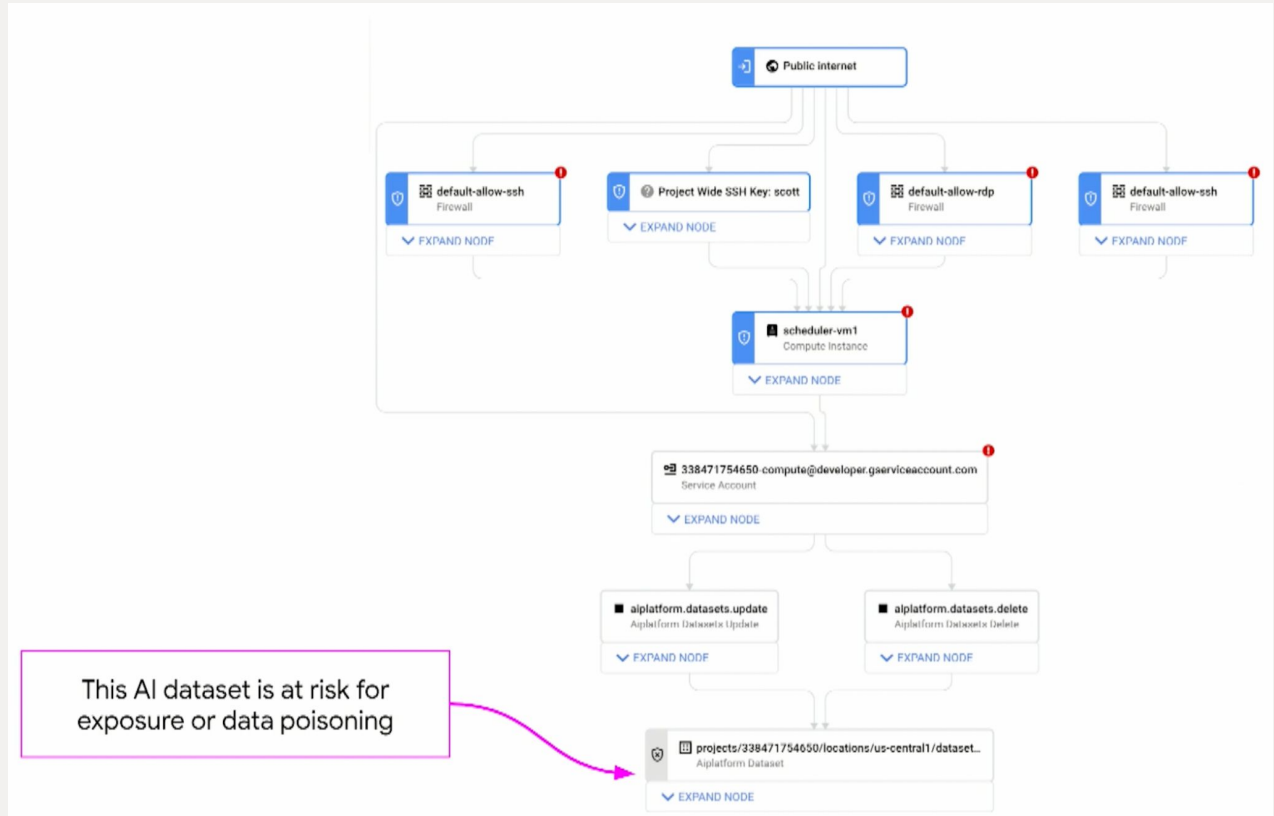
- **Google Console + CLI**
- **Sensitive Data Protection (Discovery/Profiling)**
- **Cloud Storage (GCS)**
- **BigQuery**
- **Vertex AI**

The live demo will illustrate

1. how sensitive records stored in **GCS and BigQuery** are automatically identified and tagged based on risk level.
2. We will enforce **tag-based IAM** to prevent Vertex AI from training on high-sensitivity data
3. We will see how automated **redaction workflows** can generate sanitized copies for safe model training.
4. We will also explore how Looker Studio provides an auto-generated security report summarizing sensitive asset exposure.

Understand the Data risk

Check with the AI Red Team



Fourth, access data posture and risk

Check for public exposure

Public sensitive data

TAKE ACTION10 of 12

SUMMARY

SOURCE PROPERTIES (1)

JSON

What was detected

Description

This resource contains sensitive data and can be accessed by anyone on the internet.

State

Active

Severity

High

Event time

February 4, 2025 at 4:33:28 PM GMT-8

Create time

February 4, 2025 at 4:34:58 PM GMT-8

Sensitive Data Protection data profile

[projects/dlppbqrun1-admin2s/locations/us-central1/fileStoreDataProfiles/427490147230356591](#)

cloud_dlp_data_profile data, pr

Affected resource

Resource display name

[projects/589112917390/locations/us-central1/datasets/2308730876504047616](#)

resource.display na

Resource full name

[//aiplatform.googleapis.com/projects/589112917390/locations/us-central1/datasets/2308730876504047616](#)

resource.n2

Resource type

google.aiplatform.Dataset

resource.t

Project full name

[//cloudresourcemanager.googleapis.com/projects/589112917390](#)

resource.gcp_metadata.pro

Resource path

admin > dlppbqrun1-admin2s

Cloud provider

Google Cloud

resource.cloudProv

Security contacts

None

contacts.secl

Technical contacts

None

contacts techn

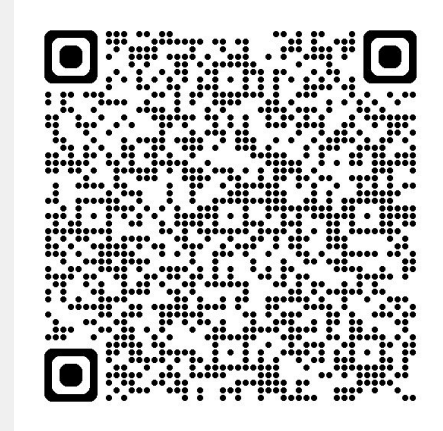
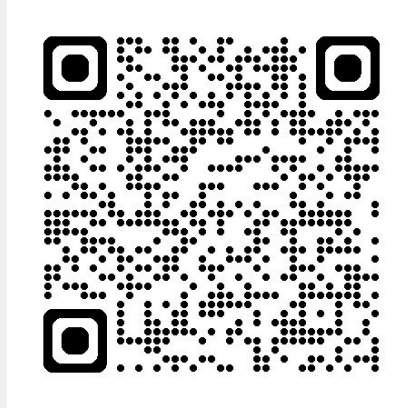
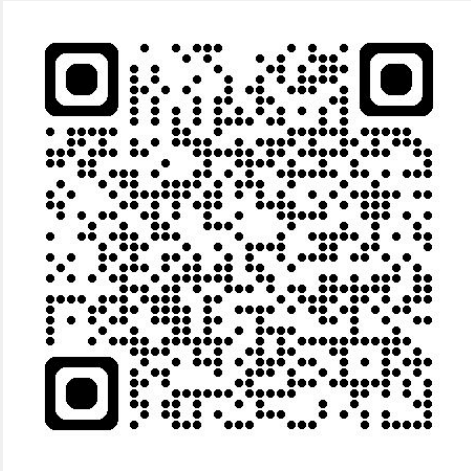
Security marks

No marks

Next steps

- To remediate public exposure, remove any IAM binding that grants all users or all authenticated users access to the related resource. To identify the related resource, click the Sensitive Data Protection data profile in this finding.
- Investigate the sensitive elements by running a Sensitive Data Protection deep inspection scan on the resource to identify all offending objects, and then remove or de-identify them.

Future References



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“Thank you to GDG Pune”



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